The empirical recurrence for OEIS A253354, proved by transfer matrix

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Abstract

OEIS A253354 counts the arrays over $\{0,\dots,1\}$ whose 2×2 subblocks carry a statistic that is monotone along named directions of the subblock grid, and carries an empirical recurrence of order 23 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. Every direction named moves the subblock row by at most one, so a pair of consecutive array rows is a state, the count is a walk count on $S = 907$ of them, and the conjectured recurrence is then settled by a finite exact computation.

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1 The conjecture

OEIS A253354 is "Number of (5+1)X(n+1) 0..1 arrays with every 2X2 subblock diagonal maximum minus antidiagonal maximum nondecreasing horizontally, vertically and ne-to-sw antidiagonally.". It has offset 1 and begins

$$907, 13917, 199512, 3013228, 44555528, 665370612, 9895936900, 147440625572, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 36*a(n-1) -406*a(n-2) +404*a(n-3) +23747*a(n-4) -145816*a(n-5) -129968*a(n-6) +3736568*a(n-7) -7902612*a(n-8) -29701104*a(n-9) +128249776*a(n-10) +34934976*a(n-11) -780114560*a(n-12) +607369728*a(n-13) +2210927104*a(n-14) -3052507136*a(n-15) -2863537152*a(n-16) +5709414400*a(n-17) +1514283008*a(n-18) -4781309952*a(n-19) -281149440*a(n-20) +1658847232*a(n-21) +94371840*a(n-22) -150994944*a(n-23) for n>26

As of the "Last modified" line on the live entry (Jul 23 2025 13:59:25, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The 2×2 subblocks of an $(n+1) \times 6$ array over $\{0,\dots,1\}$ form a grid of n rows and 5 columns; write $\sigma(i,j)$ for the value the entry's statistic takes on the subblock in position (i,j), a function of its four entries $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ alone. The condition is

$\sigma(i,j)\{le\sigma(i+1,j+0)$, $\sigma(i,j)\{le\sigma(i+0,j+1)$, $\sigma(i,j)\{ge\sigma(i+1,j-1)$

whenever both positions exist. The entry writes the array with n as the number of COLUMNS; transposing it exchanges the two coordinates of every direction, which is done once, and a direction that then points upward is turned round by reversing its inequality.

Three points of the English are fixed by the entries' own terms rather than guessed: "ne-to-sw antidiagonally" is the offset $(+1,-1)$ on the subblock grid, since the offset $(+1,+1)$ gives 164 where the corresponding entry publishes 126; "diagonal minus antidiagonal sum" is $(p+s)-(q+r)$; and "ne-sw antidiagonal difference" is $q-r$. Each was checked by counting the small arrays directly and comparing with the entry.

3 The count is a walk count

A pair of consecutive array rows determines one whole row of the subblock grid. Take those pairs as states. The comparisons with offset $(0,\pm1)$ live inside a single subblock row, so they decide which pairs are admissible at all; the comparisons with offset $(\pm1,\cdot)$ relate two consecutive subblock rows, so they decide which pair may follow which. A step of the walk appends one array row, turning the pair (r,s) into (s,t), and settles every comparison between the two subblock rows at once. An $(n+1)$-row array is therefore a walk of $n-1$ steps and every walk comes from exactly one array, so

$$a(n) \;=\; \iota^\top M^{n-1}\tau, \qquad \iota=\tau=(1,\dots,1)^\top,$$

on the $S=907$ admissible pairs. In particular a is C-finite. The alignment of the walk index with the entry's own n was checked against its published terms rather than assumed.

4 The proof

Lemma 1. *Let $q(t)=t^{23}-\sum_i c_i t^{23-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n)-\sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau=0$ for all large j, and it is enough to examine $j<S$.*

Proof. Factor M^j out of $M^{j+23}-\sum_i c_i M^{j+23-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j=\iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; 36\,a(n{-}1){-}406\,a(n{-}2){+}404\,a(n{-}3){+}23747\,a(n{-}4){-}145816\,a(n{-}5){-}129968\,a(n{-}6){+}3736568\,a(n{-}7)$$

for every $n>26$.

Proof. The vector $w=q(M)\tau$ was formed by 23 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j=0,1,\dots$, again exactly; those integers vanish from the index corresponding to $n=26+1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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